

SOFTWARE REQUIREMENTS SPECIFICATION

Project: Online Internship & Placement Preparation Portal

Version: 1.0

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1.0	06-09-2026	Project Team	Initial SRS Document	Pending

Approvals

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for the **Online Internship & Placement Preparation Portal**.

The system is designed to provide students with a single platform to search and apply for internships, attend aptitude tests, monitor preparation progress, and view performance analytics.

The system also provides facilities for recruiters to post internship opportunities and manage applications, while administrators can manage users, internships, tests, and system activities.

This document describes the functional requirements, non-functional requirements, security requirements, interfaces, system features, and acceptance criteria of the proposed system.

1.2 Scope

The Online Internship & Placement Preparation Portal provides an integrated platform for internship discovery and placement preparation.

The system includes:

- Student registration and login
- Student profile management
- Internship listing and searching
- Internship application
- Application status tracking
- Aptitude tests
- Automatic test evaluation
- Test history and performance tracking
- Placement preparation progress tracking
- Performance analytics
- Recruiter registration and login
- Internship posting and management
- Candidate application management
- Candidate shortlisting and rejection
- Administrator management of users, internships, tests and questions

The system does not directly conduct company interviews, provide employment guarantees, or perform final recruitment decisions outside the platform.



1.3 Audience

This document is intended for:

- Students
- Project developers
- Project guide
- Faculty members
- Recruiters
- System administrators
- Testers and evaluators

1.4 Definitions and Acronyms

Term	Definition
SRS	Software Requirements Specification
UI	User Interface
API	Application programming Interface
MERN	MongoDB, Express.js, React.js and Node.js
FR	Functional Requirement
NFR	Non-Functional Requirement
RTM	Requirements Traceability Matrix
OTP	One-Time Password
RBAC	Role-Based Access Control
Student	User who searches and applies for internships and takes preparation tests
Recruiter	User who posts internships and manages applications
Admin	User responsible for managing the overall system
Aptitude Test	Online test used to evaluate a student's placement preparation
Analytics	Analysis of student test performance and preparation progress

2. Overall Description

2.1 Product Perspective

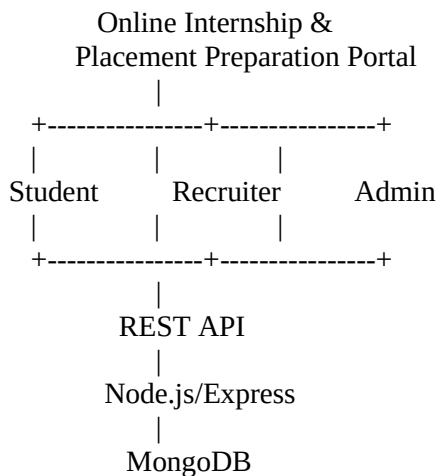
The Online Internship & Placement Preparation Portal is a web-based application developed using the MERN stack.

The system consists of:

- React.js frontend
- Node.js and Express.js backend
- MongoDB database
- REST APIs for communication between frontend and backend
- Authentication and role-based access control
- Analytics module for student performance

The system connects students, recruiters and administrators through a common platform.

Basic system architecture



2.2 Major Product Functions

The major functions of the system are:

Student Functions

- Student registration and login
- Profile management
- Internship search

- Internship filtering
- Internship application
- Application status tracking
- Aptitude test participation
- Automatic test evaluation
- Test history
- Preparation progress tracking
- Performance analytics

Recruiter Functions

- Recruiter registration and login
- Recruiter profile management
- Internship creation
- Internship editing and deletion
- Viewing applications
- Candidate shortlisting
- Candidate rejection
- Application status updates

Admin Functions

- User management
- Internship management
- Aptitude test management
- Question management
- Monitoring applications
- Viewing system-level analytics

2.3 User Roles and Characteristics

Student

Students use the system to find internship opportunities and prepare for placement examinations.

Expected characteristics:

- Basic computer knowledge
- Can register and maintain a profile
- Can search and apply for internships
- Can take aptitude tests

- Can monitor preparation progress

Recruiter

Recruiters represent companies and use the system to publish internship opportunities and manage applications.

Expected characteristics:

- Basic knowledge of recruitment activities
- Can create internship listings
- Can view candidate applications
- Can shortlist or reject candidates

Administrator

The administrator manages and monitors the overall platform.

Expected characteristics:

- Knowledge of system administration
- Can manage users and content
- Can manage aptitude tests and questions
- Can monitor platform activities

2.4 Operating Environment

The system shall operate in:

Client Side

- Desktop computers
- Laptops
- Tablets
- Smartphones
- Modern web browsers such as Chrome, Edge, Firefox and Safari

Server Side

- Node.js runtime environment
- Express.js server

- MongoDB database
- Internet connection

Development Environment

- Visual Studio Code
- Node.js
- MongoDB
- Git/GitHub
- React.js

2.5 Constraints

The system has the following constraints:

- Internet connection is required for normal operation.
- Users must provide valid registration information.
- Only authenticated users can access protected functions.
- Students can apply only for available internships.
- Recruiters can manage only their own internship listings.
- Administrators have higher-level management privileges.
- The system depends on the availability of the database and server.
- The system should follow secure coding practices for handling user information.

3. External Interface Requirements

3.1 User Interface

The system shall provide a simple and responsive web interface.

Student Interface

The student dashboard shall provide:

- Profile
- Internship listings
- Search and filter

- Applied internships
- Aptitude tests
- Test results
- Preparation progress
- Performance analytics

Recruiter Interface

The recruiter dashboard shall provide:

- Recruiter profile
- Create internship
- Manage internships
- View applications
- Candidate status management

Admin Interface

The administrator dashboard shall provide:

- User management
- Internship management
- Test management
- Question management
- System analytics

3.2 Software Interfaces

The system shall communicate between the frontend and backend using REST APIs.

Frontend

- React.js
- HTML
- CSS
- JavaScript

Backend

- Node.js

- Express.js

Database

- MongoDB

API Communication

React.js



HTTP/HTTPS Requests



Express.js / Node.js



Database Queries



MongoDB

3.3 Communication Interfaces

- The system shall use HTTP/HTTPS for client-server communication.
- HTTPS should be used for secure deployment.
- REST APIs shall be used for communication between frontend and backend.
- JSON shall be used for data exchange between frontend and backend.
- Appropriate error responses shall be returned when requests fail.

4. System Features

The following requirements define the major functional features of the system.

At least **15 functional requirements** are required. This SRS contains **18 functional requirements**.

4.1 User Registration and Authentication

Description

The system shall provide secure registration and login functionality for students and recruiters.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance / Criteria / Test Case	Dependencies
IPR-F-001	The system shall allow students to register using valid personal and academic details.	Functional	High	Student	Valid registration creates a student account. TC-Auth-01	Database
IPR-F-002	The system shall authenticate registered users using valid login credentials.	Functional	High	Student/Recruiter	Valid credentials provide access to the appropriate dashboard. TC-Auth-02	Authentication module
IPR-F-003	The system shall provide role-based access for Student, Recruiter and Admin users.	Functional	High	Admin	Users can access only their authorized modules. TC-Auth-03	Authentication

4.2 Student Profile Management

Description

Students shall be able to create and maintain their placement profile.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance / Criteria / Test Case	Dependencies
IPR-F-004	The system shall allow students to create and update their profile information.	Functional	High	Student	Updated information is displayed after saving. Test: TC-PROF-01	Student Account
IPR-F-005	The system shall allow students to maintain academic details, skills and resume information.	Functional	Medium	Student	Valid profile information can be added and updated. Test: TC-PROF-02	Student Profile

4.3 Internship Search and Application

Description

Students shall be able to discover suitable internship opportunities and submit applications.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance / Criteria / Test Case	Dependencies
IPR-F-006	The system shall display available internship opportunities to students.	Functional	High	Student	Available internships are displayed correctly. Test: TC-INT-01	Internship database
IPR-F-007	The system shall allow students to search and filter internships by relevant criteria.	Functional	High	Student	Search/filter returns matching internships. Test: TC-INT-02	Internship listings
IPR-F-008	The system shall allow students to apply for an available internship.	Functional	High	Student	Successful application is stored in the database. Test: TC-INT-03	Student profile
IPR-F-009	The system shall prevent a student from submitting multiple applications	Functional	High	Student	Duplicate application is rejected. Test: TC-INT-04	Application database

	for the same internship.					
IPR-F-0010	The system shall allow students to view the current status of their internship applications.	Functional	High	Student	Student can view submitted, shortlisted, rejected or other configured statuses. Test: TC-INT-05	Application module

4.4 Aptitude Test Module

Description

The system shall provide online aptitude tests for placement preparation.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance / Criteria / Test Case	Dependencies
IPR-F-011	The system shall allow students to participate in available aptitude tests.	Functional	High	Student	Student can start and complete an available test. Test: TC-TEST-01	Test module

IPR-F-012	The system shall calculate the student's score automatically after test submission.	Functional	High	Student	Score is calculated correctly based on submitted answers. Test: TC-TEST-02	Question database
IPR-F-013	The system shall store the student's test scores and test history.	Functional	High	Student	Completed test results are stored and displayed. Test: TC-TEST-03	Database

4.5 Placement Preparation Progress

Description

The system shall help students monitor their preparation progress.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance / Criteria / Test Case	Dependencies
IPR-F-014	The system shall allow students to record and update placement preparation progress.	Functional	Medium	Student	Progress updates are saved successfully. Test: TC-PROG-01	Student profile

IPR-F-015	The system shall display performance analytics based on aptitude test results and preparation progress.	Functional	High	Student	Analytics are generated from available student data. Test: TC-PROG-02	Test and progress modules
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4.6 Recruiter Module

Description

Recruiters shall be able to manage internship opportunities and applications.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case	Dependencies	
IPR-F-016	The system shall allow recruiters to create, update and manage internship listings.	Functional	High	Recruiter	Recruiter can create and modify their internship listings. Test: TC-REC-01	Recruiter account	
IPR-F-017	The system shall allow recruiters to view and update internship applications.	Functional	High	Recruiter	Recruiter can view applications and update candidate status. Test: TC-REC-02	Application module	

4.7 Administration Module

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case	Dependencies	
IPR-F-018	The system shall allow administrators to manage users, internships, aptitude tests and questions.	Functional	High	Admin	Admin can add, update or remove permitted system data. Test: TC-ADMIN-01	Admin authentication	

5. Non-Functional Requirements

The following requirements define the quality and operational characteristics of the system.

At least **5 NFRs** are required. This SRS contains **8 NFRs**.

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
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IPR-NF-001	The main dashboard shall load within 3 seconds under normal network and server conditions.	Performance	High	Response time measured using performance testing. Test: TC-PERF-01
IPR-NF-002	The system shall provide at least 99% availability during normal operating periods.	Reliability	High	Availability monitored during testing/deployment. Test: TC-REL-01
IPR-NF-003	The system shall support at least 500 concurrent users under the planned deployment configuration.	Scalability	Medium	Load testing shall be performed with simulated concurrent users. Test: TC-SCALE-01
IPR-NF-004	The user interface shall be responsive on desktop, tablet and mobile screen sizes.	Usability	High	UI tested on different screen sizes. Test: TC-UI-01
IPR-NF-005	The system shall validate user inputs before storing them in the database.	Security/Reliability	High	Invalid inputs are rejected. Test: TC-SEC-01
IPR-NF-006	The system shall maintain	Data Integrity	High	Database records remain consistent

	consistent application, test and user records.			after operations. Test: TC-DATA-01
IPR-NF-007	The system shall be maintainable using modular frontend and backend components.	Maintainability	Medium	Modules can be modified without affecting unrelated modules. Test: TC-MAIN-01
IPR-NF-008	The system shall provide clear error messages when an operation fails.	Usability	Medium	User receives understandable error feedback. Test: TC-UX-01

6. Quality Attributes & Acceptance Tests

6.1 Quality Attributes

The system shall demonstrate the following quality attributes:

Performance

The system should respond quickly to user requests and provide smooth navigation.

Reliability

User applications, test results and progress records should remain consistent and available.

6.2 Acceptance Test Criteria

The project shall be considered acceptable when:

1. All high-priority functional requirements are implemented.
2. Students can register and log in successfully.
3. Students can search and apply for internships.
4. Students can view their application status.
5. Students can take aptitude tests.
6. Test scores are calculated correctly.
7. Student preparation progress and analytics are displayed.
8. Recruiters can create and manage internship listings.
9. Recruiters can view and update applications.
10. Administrators can manage users, internships and tests.
11. Security requirements are verified.
12. No critical functional or security defects remain.
13. RTM shows that all major requirements are mapped to test cases.

7. System Models and UML Diagrams

At least **2 UML Use-Case Diagrams** shall be prepared.

7.1 UML Use-Case Diagram 1 – Student Module

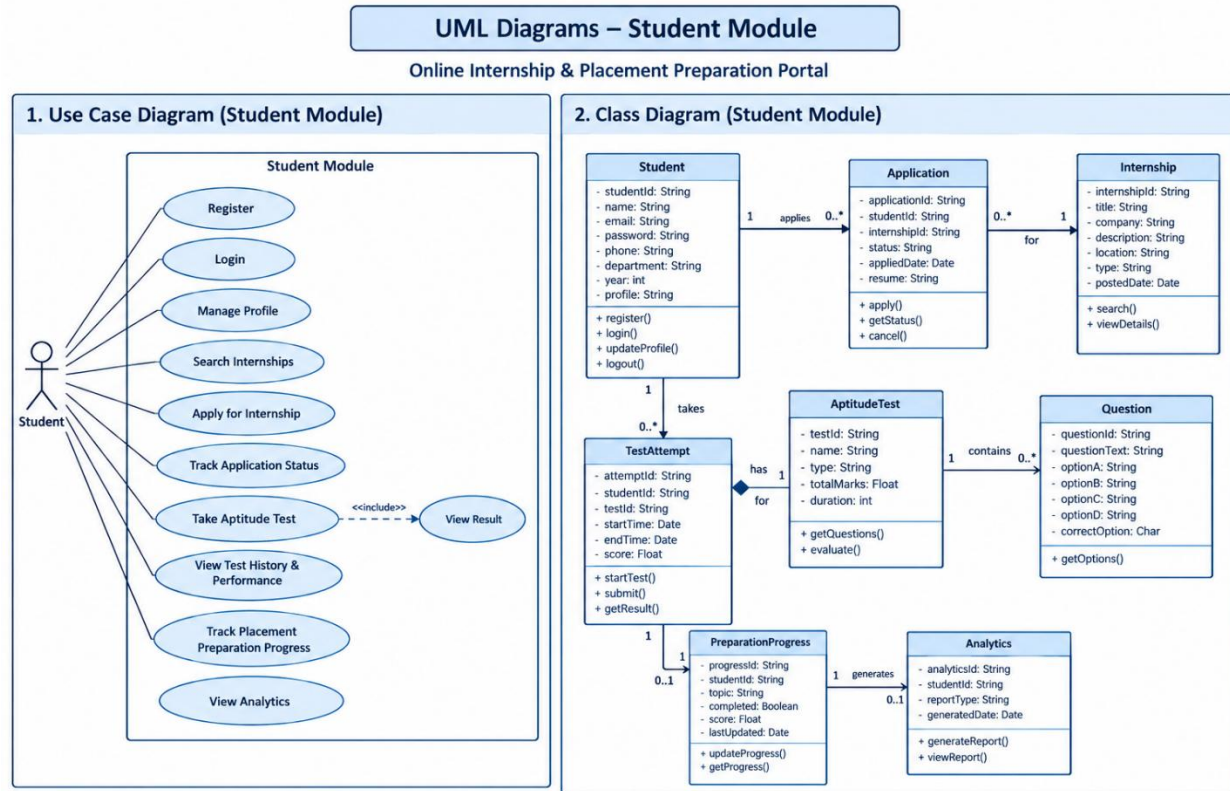
Actor

Student

Use Cases

- Register
- Login
- Manage Profile
- Browse Internships
- Search/Filter Internships
- View Internship Details
- Apply for Internship
- Track Application
- Take Aptitude Test
- View Test Result
- View Test History
- Update Preparation Progress
- View Performance Analytics
- Logout

Suggested relationship



7.2 UML Use-Case Diagram 2 – Recruiter and Admin Module

Actors

- Recruiter
- Admin

Recruiter Use Cases

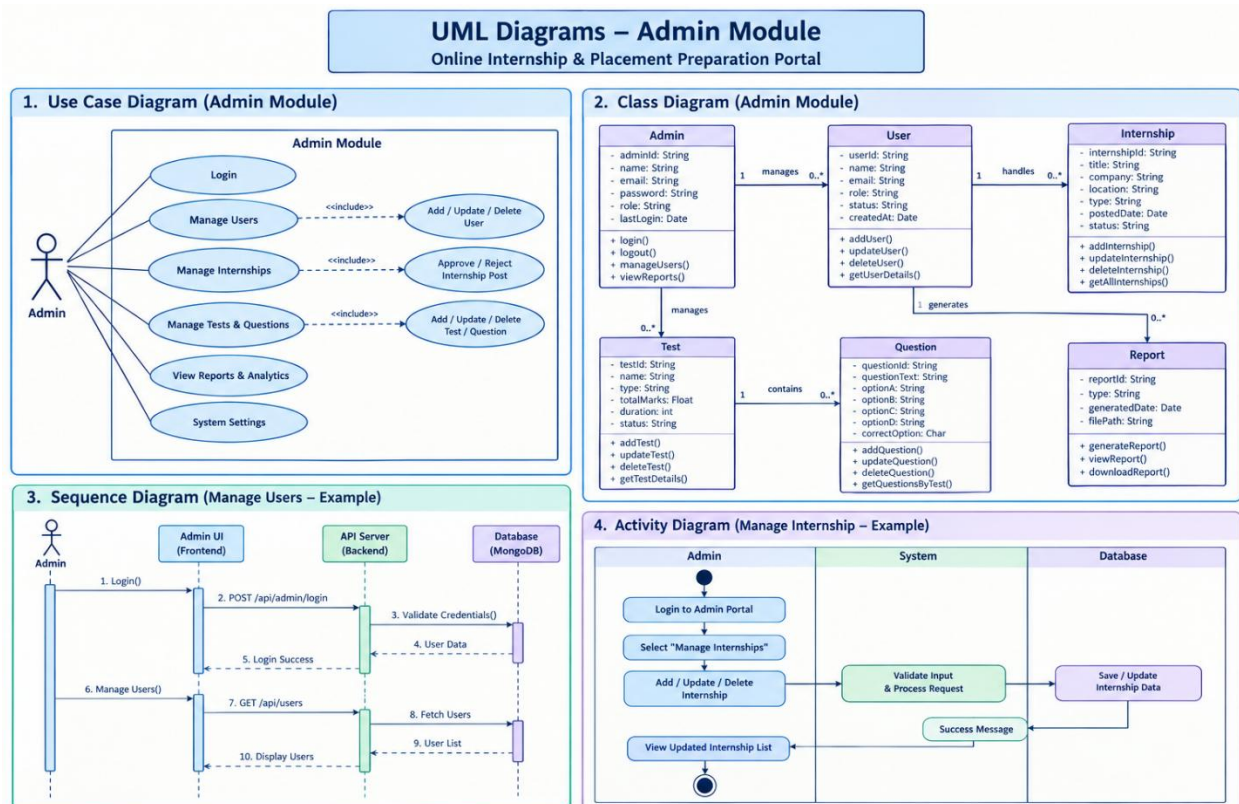
- Register/Login
- Manage Recruiter Profile
- Create Internship
- Update Internship
- Delete/Close Internship
- View Applications

- Shortlist Candidate
- Reject Candidate
- Update Application Status

Admin Use Cases

- Login
- Manage Students
- Manage Recruiters
- Manage Internships
- Manage Aptitude Tests
- Manage Questions
- Monitor Applications
- View System Analytics

Suggested structure



8. Requirements Traceability Matrix (RTM)

The RTM maps requirements to their corresponding modules and test cases.

Status:

- N = Not Started
- P = Partially Implemented
- A = Approved/Completed

Req ID	Requirement Short	Section / Design Spec	Module	Test Case(s)	Status
IPR-F-001	Student registration	4.1 / DS-AUTH-01	Authentication	TC-AUTH-01	N
IPR-F-002	User login	4.1 / DS-AUTH-02	Authentication	TC-AUTH-02	N
IPR-F-003	Role-based access	4.1 / DS-AUTH-03	Authentication	TC-AUTH-03	N
IPR-F-004	Student profile	4.2 / DS-PROF-01	Student Profile	TC-PROF-01	N
IPR-F-005	Academic/skill details	4.2 / DS-PROF-02	Student Profile	TC-PROF-02	N
IPR-F-006	Internship listings	4.3 / DS-INT-01	Internship	TC-INT-01	N
IPR-F-007	Search/filter	4.3 / DS-INT-02	Internship	TC-INT-02	N
IPR-F-008	Apply internship	4.3 / DS-INT-03	Application	TC-INT-03	N
IPR-F-009	Prevent duplicate application	4.3 / DS-INT-04	Application	TC-INT-04	N
IPR-F-010	Track application	4.3 / DS-INT-05	Application	TC-INT-05	N
IPR-F-011	Take aptitude test	4.4 / DS-TEST-01	Aptitude Test	TC-TEST-01	N
IPR-F-012	Calculate test score	4.4 / DS-TEST-02	Aptitude Test	TC-TEST-02	N

IPR-F-013	Store test history	4.4 / DS-TEST-03	Aptitude Test	TC-TEST-03	N
IPR-F-014	Preparation progress	4.5 / DS-PROG-01	Progress	TC-PROG-01	N
IPR-F-015	Performance analytics	4.5 / DS-PROG-02	Analytics	TC-PROG-02	N
IPR-F-016	Manage internships	4.6 / DS-REC-01	Recruiter	TC-REC-01	N
IPR-F-017	Manage applications	4.6 / DS-REC-02	Recruiter	TC-REC-02	N
IPR-F-018	Admin management	4.7 / DS-ADMIN-01	Admin	TC-ADMIN-01	N
IPR-NF-001	Response time	5 / DS-PERF-01	Web/API	TC-PERF-01	N
IPR-NF-002	Availability	5 / DS-REL-01	System	TC-REL-01	N
IPR-NF-003	Concurrent users	5 / DS-SCALE-01	System	TC-SCALE-01	N
IPR-NF-004	Responsive UI	5 / DS-UI-01	Frontend	TC-UI-01	N
IPR-NF-005	Input validation	5 / DS-SEC-01	Backend	TC-SEC-01	N
IPR-NF-006	Data integrity	5 / DS-DATA-01	Database	TC-DATA-01	N
IPR-NF-007	Maintainability	5 / DS-MAIN-01	All Modules	TC-MAIN-01	N
IPR-NF-008	Error handling	5 / DS-UX-01	Frontend/API	TC-UX-01	N
IPR-SR-001	Authentication protection	6.2 / DS-SEC-01	Authentication	TC-SEC-01	N
IPR-SR-002	Password hashing	6.2 / DS-SEC-02	Authentication	TC-SEC-02	N
IPR-SR-003	Role authorization	6.2 / DS-SEC-03	Authorization	TC-SEC-03	N
IPR-SR-004	HTTPS/TLS	6.2 / DS-SEC-04	Communication	TC-SEC-04	N
IPR-SR-005	Input sanitization	6.2 / DS-SEC-05	Backend	TC-SEC-05	N

9. Summary

The **Online Internship & Placement Preparation Portal** provides an integrated platform for internship searching, application management and placement preparation.

The system supports three main user roles: **Student, Recruiter and Admin.**

Students can search and apply for internships, attend aptitude tests, track their preparation progress and view performance analytics. Recruiters can publish internship opportunities and manage applications. Administrators can manage users, internships, tests and questions.

The proposed system contains:

- **18 Functional Requirements**
- **8 Non-Functional Requirements**
- **2 Security Objectives**
- **5 Security Requirements**
- **2 UML Use-Case Diagrams**
- **Complete Requirements Traceability Matrix**

The system is planned using the **MERN stack: MongoDB, Express.js, React.js and Node.js**, with REST APIs connecting the frontend and backend.

Project Contribution Details

S. No.	Name of the Student	SRN	Contribution
1	Nikhil Raj G	PES2UG24CS319	Introduction and Overall Description
2	NAVYA	PES2UG24CS308	External Interface Requirements and System Features
3	Patnaikuni Sai Dheeraj	PES2UG24CS341	Non-Functional Requirements and Quality Attributes & Acceptance Tests
4	Sohel M Chapparband	PES2UG25CS826	UML Use-Case Diagrams and Requirements Traceability Matrix (RTM)